

Mohamed Mousa, Ph.D.

Computational Neuroscientist

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[LinkedIn](#)



[GitHub](#)



[Google Scholar](#)



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SUMMARY

Neuroengineer with 8+ years of experience in computational neuroscience, electrophysiology, neural signal processing, and biophysical modeling. Developed NEURON-based models of spinal circuits and electrical stimulation effects, built scalable Python/MATLAB pipelines for neural data analysis, and conducted human and animal electrophysiology experiments. Experience spanning neural decoding, EMG analysis, nonlinear neural dynamics, and machine learning for neural systems.

EXPERIENCE

Postdoctoral Fellow Research Associate

Oct 2024 – Present

Temple University — Philadelphia, PA

- Built end-to-end pipelines to process, decompose, and analyze electrophysiological data (128 channel) using MATLAB, Python, and R.
- Developed Python-based analysis pipelines to quantify functional connectivity and nonlinear dynamics in spinal interneuron networks using graph-theoretic and statistical methods.
- Conducted high-density electrode recordings from human subjects and animal models across multiple experimental conditions.

Postdoctoral Researcher

Jan 2023 – Oct 2024

Wright State University — Dayton, OH

- Led IRB-approved clinical study to develop a diagnostic biomarker for ALS — enrolled, consented, and performed EMG recordings on human participants (20+).
- Analyzed and reverse-engineered electrophysiological recordings from human and animal models to identify disease signatures.
- Developed predictive biophysical computational models using NEURON and MATLAB to classify ALS-related neural changes.

Doctoral Candidate

Jun 2017 – Dec 2022

Wright State University — Dayton, OH

- Developed 3D models of motoneurons to reverse-engineer experimental electrophysiological data.
- Modeled electrical field potentials and the effects of neural stimulation on motoneuron excitability.
- Automated analysis and feature extraction of electrophysiological datasets using Python and MATLAB, reducing analysis time by 60×.
- Designed mathematical models of ionic channels to improve computational accuracy and replicate complex in-silico motoneuron behaviors.

Research Assistant

Dec 2014 – Jun 2017

Cairo University — Cairo, Egypt

- Taught courses in medical equipment, bio-electronics, data structures, and computer vision.
- Supervised BCI graduation projects using P300 brain waves.

Field Clinical Engineer

2013 – 2014

Maadi Military Hospital – Cairo, Egypt

- Supported installation, calibration, and maintenance of MRI, CT, and ventilator systems in clinical environments, coordinating directly with physicians and hospital engineering staff.

EDUCATION

Ph.D. in Engineering — Computational Neuroscience 2022

Wright State University, Dayton, OH

Dissertation: *Reverse Engineering Spinal Motoneuron Properties in Slice and Whole Cord Preparations Using Computational Modeling*

SELECTED PUBLICATIONS

Full publication list available on [Google Scholar](#).

- Gamal M, Mousa MH, Eldawlatly S, Elbasiouny SM (2021) In-silico development and assessment of a Kalman filter motor decoder for prosthetic hand control. *Comput Biol Med.* 132, 104353
- Mousa M. et al. Onion skin is not a universal firing pattern for spinal motoneurons: simulation study. *Journal of neurophysiology*, 132(1), 240-258.
- Mahrous AA, Mousa MH, Elbasiouny SM (2019) The Mechanistic Basis for Successful Spinal Cord Stimulation to Generate Steady Motor Outputs. *Frontiers in Cellular Neuroscience* 13:359-374

TECHNICAL SKILLS

Domain Expertise

Machine Learning · Computational Modeling & Simulation · Signal & Image Processing · Network & Graph Theory · Statistical Analysis · Nonlinear Dynamics · Clinical Management & IRB Protocols, Parallel Computing

Programming Languages

Python · MATLAB · R · C/C++ · JavaScript · NEURON (Hoc, NMODL) · SQL

Libraries & Tools

NumPy · SciPy · Pandas · Matplotlib · Seaborn · Plotly · PyTorch · statsmodels · Pingouin · SAS

AWARDS

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| • Excellence award as the top Ph.D. graduate student in biomedical engineering, WSU, USA | 2023 |
| • Science Gateways Community Institute (SGCI) Science Ambassador - \$1200 Travel award | 2019 |
| • Travel Award by Professional Development Grant, WSU, USA | 2018 |
| • Best paper award, Cairo International Biomedical Engineering (CIBEC) | 2016 |
| • Graduated First in Class with Distinction, Cairo University, Egypt | 2013 |
| • Recipient of annual Excellence Award (top 5%), Cairo University, Egypt | 2009-2012 |